



Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

College of Computer Studies

ITEC 204 – Data Structures and Algorithms

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Section: BSIT-2B	Instructor: MS. Ignacio

- I. **Objective:** Write a short explanation of what your program is designed to do. (Minimum of 3 sentences)

Answer:

This program is a simple incident ticket tracker for monitoring automated bots that have failed or encountered errors. It allows the user to add new incident tickets by entering an Incident ID, the affected Bot's name, and a short description of the issue, storing each ticket as a record in a list. The user can also display all currently active tickets, search for a specific ticket using its Incident ID, and remove a ticket once it has been resolved. Finally, the program can display the total number of active incident tickets at any time, giving the user a quick overview of how many issues are still outstanding. The program runs continuously through a menu-driven loop until the user chooses to exit, making it easy to perform multiple operations in a single session.

- II. **Program Operations:** Briefly explain how your program performs each operation. (Insert Specific Code Line and Your Explanation)

Add Incident:

```
while True:
    print("\n1. Add \n2. Display \n3. Search \n4. Remove \n5. Count \n6. Exit")
    choice = input("Choice: ")

    if choice == "1":
        new_id = input("Incident ID: ")
        bot = input("Bot: ")
        desc = input("Short Description: ")
        incident.append([new_id, bot, desc])
        print("Ticket added.")
```

In this program, the program shows a menu that if the user chose option 1 or add ticket option, it will ask the user to input a new incident id, bot, and description.



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Display Incidents:

```
elif choice == "2":  
    if not incident:  
        print("No active incident tickets.")  
    else:  
        for t in incident:  
            print(t[0], "|", t[1], "|", t[2])
```

In this program, if the user chooses option 2 or display tickets options, it will analyze the program to see if there's an active ticket. If no tickets are active, it will show a message that says, "No active incident tickets". In this case, since we already have 10 incident tickets, it will show the 10 sample tickets.

Search Incident:

```
elif choice == "3":  
    search_id = input("Incident ID: ")  
    for t in incident:  
        if t[0] == search_id:  
            print("Found:", t[0], "|", t[1], "|", t[2])  
            break  
    else:  
        print("Not found.")
```

In this program, it will ask you to input the incident id and if no incident id matched, it will show a "Not found" message, but if an incident matched, it will show the incident id, bot, and the description

Remove Resolved Incident:

```
elif choice == "4":  
    remove_id = input("Incident ID to remove: ")  
    for i in range(len(incident)):  
        if incident[i][0] == remove_id:  
            incident.pop(i)  
            print("Ticket removed.")  
            break  
    else:  
        print("Not found.")
```

In this lines of code, it will ask the user to input the incident id and if no incident id matched, it will show a "Not found" message and if it a incident id matched, it removes the ticket that has the entered incident id and it will show a "Ticket removed." message.

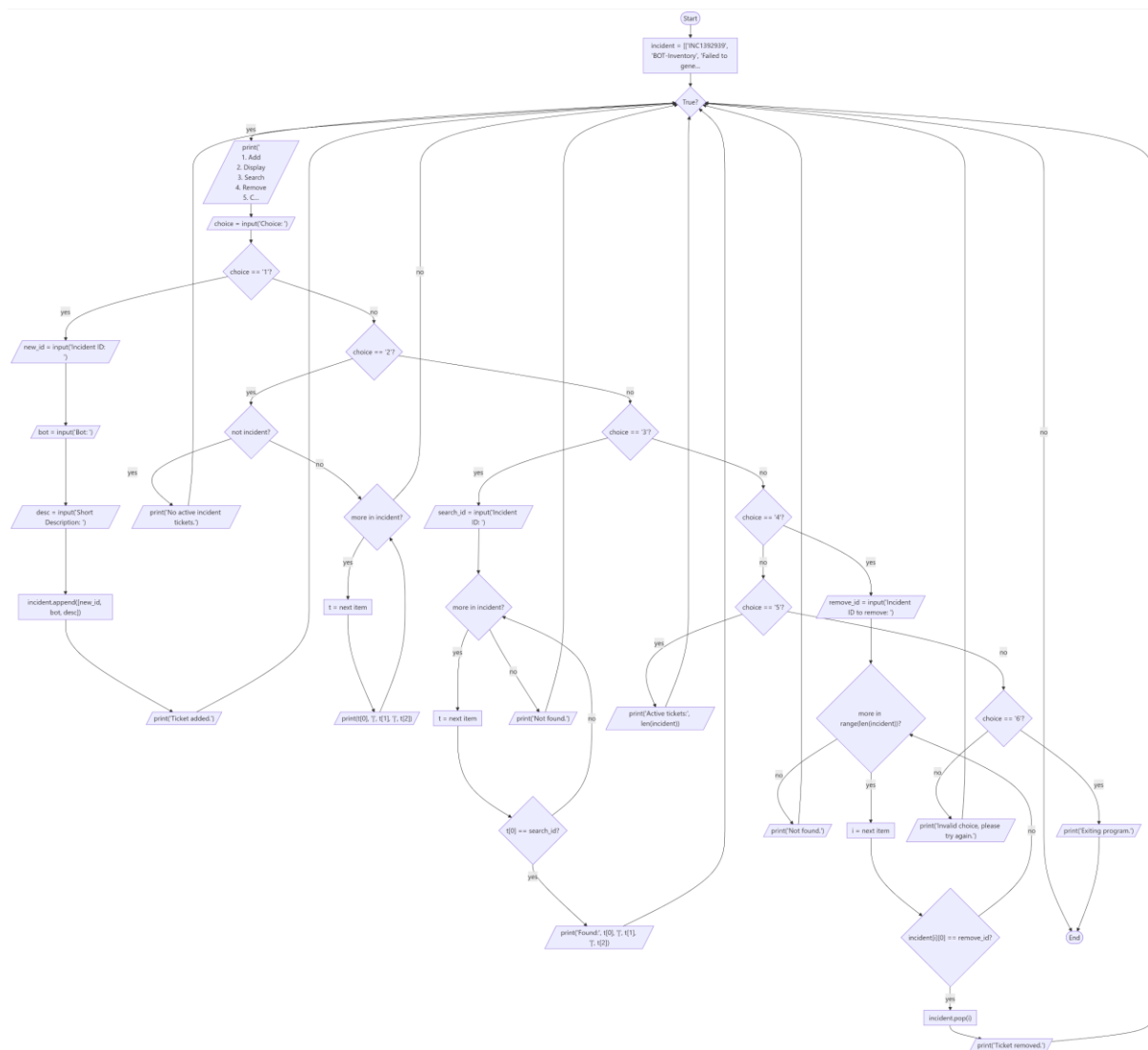


Count Active Incidents:

```
elif choice == "5":  
    print("Active tickets:", len(incident))
```

In this code, it just count the active tickets.

III. Flowchart: Create a flowchart showing the overall process and logic of your program.





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- IV. **Conclusion:** Briefly explain what you learned from the activity and how the linear data structure was applied.

From doing this activity, I learned how to use a list of lists in Python to store and manage records, kind of like a simple table with rows and columns. Each incident ticket was stored as a small list with its ID, Bot name, and description inside it, and I could get each piece of info using index numbers like `t[0]`, `t[1]`, and `t[2]`. I also learned how to use loops to go through the list to search for a ticket or remove one, and how `append()` and `pop()` can be used to add and delete items from a list. This activity helped me understand how a linear data structure like a list can be used to organize and manage data in a program, especially when you need to add, search, and remove information one at a time.
